MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIAL BEAM SWITCHING

TYPE 6703

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: External permanent magnet, magnetically shielded, high vacuum, 10 outputs

DIMENSIONS AND PIN CONNECTIONS: See figure 4

ABSOLUTE RATINGS:

Parameter:	Ef	Eta	Esp(Note 8)	Ecc	Rta(Note 8)	Rsp(Note 8)	Esig
Unit:	V	Vdc	Vdc	Vdc	Ohms	Ohms	V
Maximum:	6.9	300	200	200	---	---	---
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS: (Note 1)	6.3	100	100	25	3.3K ±1% (Note 12)	100K ±1%	-65

ABSOLUTE RATINGS:

Parameter:	Ehk	F	TA	P/ta	Ik	Alt
Unit:	V	MHz	°C	W	mA	Ft
Maximum:	+100	Note 7	+85	1	20	60,000
Minimum:	-100	Static	-65	---	---	---
TEST CONDITIONS: (Note 1)	---	---	---	---	---	---

GENERAL:

(B) Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMIT MIN	LIMIT MAX	UNIT
	Quality conformance inspection, part 1								
---	Counting voltage range (1)	3, 4, 5, 9, 14	4 Hz	0.65	II	Eta / Esp	85	115	Vdc
---	Counting voltage range (2)	3, 4, 5, 9, 14	10 kHz	0.65	II	Eta / Esp	85	115	Vdc
1256	Electrode current (target)	-	Individual targets	0.65	II	Ita	5.5	7.5	mA
	Quality conformance inspection, part 2								
---	Firing voltage	17	Eta = Esp = 200 Vdc; Esig = 0	2.5	II	---	---	---	---
---	Counting voltage range (3)	3, 5, 9, 13, 14	1 MHz; Esig = -80 V	---	---	Eta / Esp	85	115	Vdc
1003	Barometric pressure, reduced	-	55 ±5 mmHg; 300 Vdc between targets and all other elements	---	---	---	---	---	---
1301	Heater current	-		---	---	If	270	330	mA
1336	Heater - cathode leakage	-		---	---	Ihk	---	20	uAdc
---	Shock	10, 15	350G; 1 ±0.5 ms duration	---	---	---	---	---	---
1031	Variable - frequency vibration	1, 4, 6, 10	Tube operating at 10 kHz	---	---	---	---	---	---
---	Shock and variable-frequency vibration end point	11		---	---	---	---	---	---
---	Magnet and shield secureness	16		---	---	---	---	---	---
2126	Envelope strain	-		---	---	---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test (1) (dynamic)	-	Group C; 120 counts/sec sine wave excitation. t = 1,000 hours (min)	---	---	---	---	---	---
---	Life test (2) (static)	-	Group C; beam formed and locked on any one target; Esig = 0; t = 1,000 hours (min)	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test end points (1) and (2):								
1256	Electrode current (target)	-	Individual targets	---	---	Ita	5.0	8.0	mA
---	Counting voltage range (1)	-		---	---	Eta | Esp	90	110	Vdc
---	Counting voltage range (2)	-		---	---	Eta | Esp	90	110	Vdc
(8) ---	Electrical noise	2, 9, 18	Individual targets	---	---	---	---	500	mVac

NOTES:

1. In order to avoid possible operational changes in the characteristics of the tube, a minimum distance of 1 inch shall be provided between ends of the magnetic surfaces when the tubes are placed end to end. No spacing is required when the tubes are placed side to side.
2. Noise is measured across Rta with an ac Electronic Voltmeter (Ballantine Model No. 300, or equivalent).
3. Tube shall count 10 positions at specified frequency and throughout counting voltage range.
4. Grid swing from +25 to -40 V with respect to cathode. Only one set of grids shall be negative with respect to the grid bias (Ecc) at any time. See figure 2 for appropriate wave shape.
5. Individual spade stem leads capacity should not exceed 10 pF with tube in socket.
6. Tube shall count 10 at specified frequency.
7. Tube is tested up to 1 MHz, but may operate to 2.0 MHz under optimum circuit conditions.
8. Following are the definitions of symbols used in this tube specification sheet:

   Rta - Resistance in series with target.
   Rsp - Resistance in series with spade.
   Esp - Spade voltage.

9. Test in accordance with circuit (see figure 1).
10. Tap short test does not apply.
11. The tube shall count 10 at the specified test conditions at 10 kHz.
12. This is the total resistance in series with each of the 10 targets and 10 spades.
13. Grid swing from +25 to -55 V with respect to cathode. Only one set of grids shall be negative with respect to the grid bias (Ecc) at any time. See figure 3 for appropriate wave shape.
14. Counting voltage ranges should be read from center point down for minimum limit, and center point up for maximum limit.
15. Shock test shall be made on any shock machine capable of producing a half-sine wave shock waveform of the specified duration and amplitude. No voltages will be applied during this test. Each tube shall be subjected to a total of 20 shocks, that is, 5 shocks in each of positions X, Y, Z plus, and Z minus in any sequence. This is a destructive test. Tubes subjected to destructive tests shall not be delivered on the contract or order.
16. Thrust between glass envelope and shield = 30 pounds minimum. The tube shall be immersed in water at 50°C for 18 hours, and then removed and cooled for 1 hour at room temperature before testing.
17. Opening the switch (SW1) in figure 1 for a minimum of 1 millisecond should clear the beam; closing switch (SW1) should not reform the bear
18. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Air Force - 11

Preparing activity:
Navy - EC

Agent:
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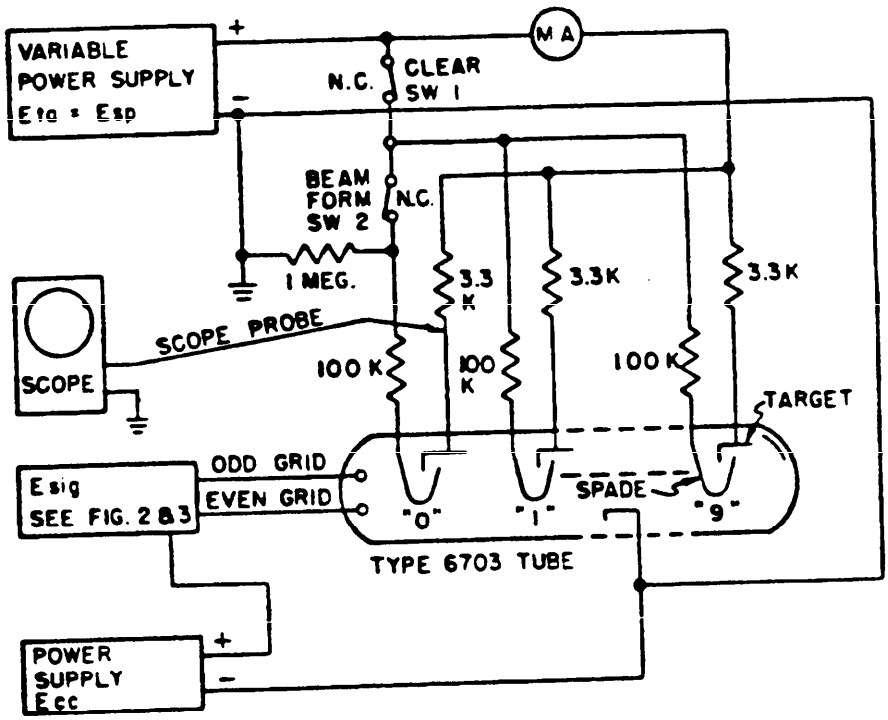


FIGURE 1. Test circuit.

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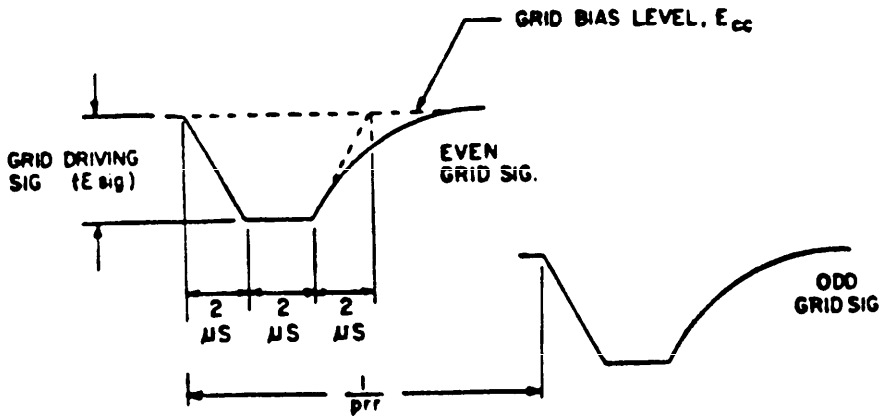


FIGURE 2. Grid signal for counting voltage range (1) and (2).

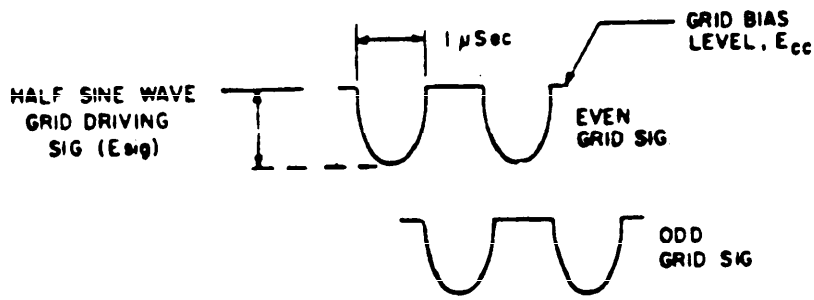
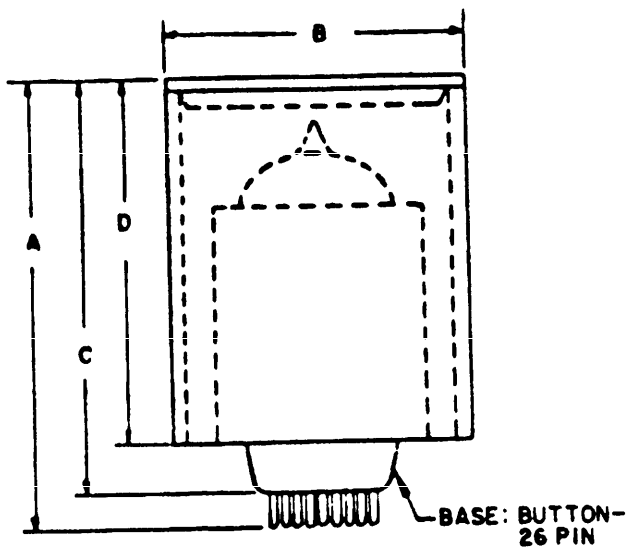


FIGURE 3. Grid signal for counting voltage range (3).

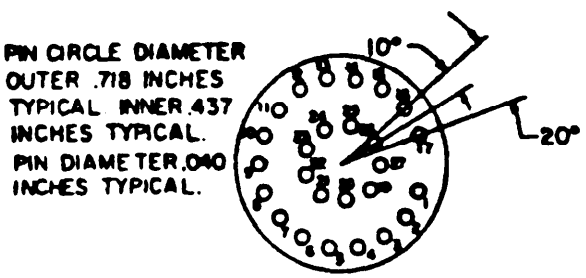
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	3.050 (77.47)	3.490 (88.65)
B	2.240 (56.90)	2.270 (57.66)
C	2.800 (71.12)	3.233 (82.12)
Reference dimensions for information only		
D	2.650 (67.31)	

FIGURE 4. Outline drawing of electron tube type 6703.

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BOTTOM VIEW

Pin No.	Element	Pin No.	Element
1	Spade-0	14	Spade - 2
2	Target - 9	15	Target - 1
3	Target - 8	16	Grid Even
4	Grid Odd	17	Target - 0
5	Target - 7	19	Spade - 9
6	Spade - 7	20	Spade - 8
7	Target - 6	21	Heater
8	Target - 5	22	Spade - 6
9	Spade - 5	23	Spade - 4
10	Target - 4	24	Spade - 3
11	Do not use	25	Heater
12	Target - 3	26	Spade - 1
13	Target - 2	27	Cathode

FIGURE 4. Outline drawing of electron tube type 6703 - Continued.

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